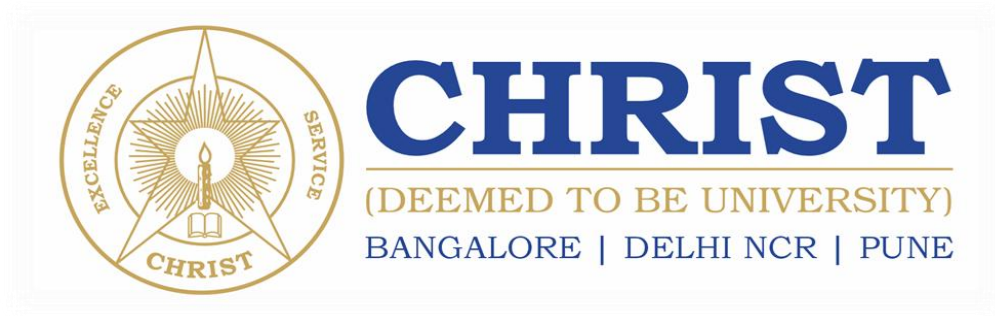




DEPARTMENT OF AI, ML & DATA SCIENCE

MICSAI785 - Capstone Project

SMART GLOVE FOR E-WASTE WORKERS

B. Tech – Department of Mechanical Engineering

With

Minors in Artificial Intelligence

**School of Engineering and Technology,
CHRIST (Deemed to be University),
Kumbalagodu, Bengaluru-560 074**

September 2025



Certificate

*This is to certify that **Abraham Thomas Joy** has successfully completed the project work for **MICSAI785 – Capstone Project** titled **Smart Glove for e-waste workers** in partial fulfillment for the award of Bachelor of Technology in Department of Electrical and Electronics Engineering with the **Minors in Artificial Intelligence** during the year 2025-2026.*

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Abstract

This report presents the design, development, and preliminary validation of an AI-enabled smart glove system aimed at enhancing the safety, comfort, and hygiene of e-waste workers exposed to hazardous and high-temperature environments. The glove integrates temperature and humidity sensors embedded within, coupled with a passive Phase Change Material (PCM) cooling mechanism, to provide continuous real-time monitoring of the glove's internal microclimate. Wireless connectivity via an ESP32 microcontroller and MQTT protocol facilitates seamless data transmission to cloud-based platforms, enabling remote monitoring and health surveillance.

The system's architecture is supported by an IoT data dashboard for live visualization and a Python-based preliminary analytics pipeline that flags environmental threshold breaches and recommends timely worker alerts. The incorporation of explainable AI methodologies, including SHAP and LIME, provides transparency to predictive heat stress notifications and ensures user confidence in AI-driven safety decisions. Digital ergonomic modeling optimizes PCM pouch placement for maximal thermal protection, while hardware prototypes and firmware code validate sensor functionality and reliable communication under typical workplace conditions.

Overall, the project demonstrates a significant step toward intelligent personal protective equipment tailored for informal e-waste sectors, addressing critical health risks such as heat stress and infectious skin conditions via integrated sensing, passive cooling, and AI-powered analytics. Future work is envisioned to expand AI capabilities for predictive hazard detection, improve sensor durability, automate sanitization workflows, and broaden applicability across related industrial sectors to advance sustainable, data-driven occupational health solutions.

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Introduction

E-waste workers operate in challenging environments where prolonged use of protective gloves is essential but problematic. Traditional gloves used in these settings often fail to address internal microclimate issues, as they tend to trap heat and sweat inside. This creates an uncomfortable environment for the workers' hands, leading to a build-up of moisture and elevated temperatures. The accumulation of heat and sweat inside the gloves results in painful skin conditions such as rashes and infections. Moreover, the persistent discomfort often causes fatigue among the workers, which significantly affects their overall work efficiency and productivity.

Compounding the issue, there is currently no integrated system that can alert the workers when the internal conditions of their gloves become hazardous. Without timely warnings, workers might unknowingly continue to use gloves that are overly hot or damp, exacerbating health risks. As a result, gloves are often reused in unsanitary conditions, increasing the likelihood of bacterial and fungal infections. This situation presents a pressing occupational safety and health concern for workers in e-waste processing, as these conditions can lead to chronic ailments and dangerous exposures.

The glove is embedded with temperature and moisture sensors that continuously monitor the internal glove environment in real time. Using passive Phase Change Material (PCM) cooling technology, the system helps regulate the temperature within the glove without relying on power-intensive active cooling methods. The inclusion of Wi-Fi connectivity enables the glove to send real-time alerts to the user and supervisors, warning them promptly if conditions reach dangerous levels and thus allowing timely interventions.

This smart glove innovation aims to significantly improve worker safety and hygiene while also extending the practical lifespan of gloves through enhanced cooling and sanitization processes. By delivering real-time feedback and environmental control, the system reduces heat stress and lowers infection risks associated with glove use in harsh e-waste environments. Ultimately, this approach promotes healthier working conditions, boosts productivity by reducing downtime caused by discomfort and illness, and contributes to sustainable glove usage in e-waste recycling operations.

Literature Review

Gu et al. present a novel design of wireless smart gloves using ultra-stable, recyclable liquid metal-based fibers, aimed at high-precision hand gesture recognition. Their work innovates by introducing sustainable, flexible sensing materials that maintain sensitivity and durability in dynamic hand movements. The glove architecture integrates these fibers seamlessly without sacrificing wearer comfort, highlighting ecological considerations alongside functional performance. This study advances the state of wearable gloves in both materials' science and human-machine interaction fields, suggesting promising applications in robotics and virtual reality.[1]

Iqbal et al. develop an affordable smart glove equipped with multiple sensors for monitoring vital health parameters non-invasively. The glove captures physiological data such as temperature, heart rate, and motion, making healthcare more accessible outside clinical environments. The design emphasizes simplicity, cost-effectiveness, and user comfort, enabling continuous health monitoring. This work contributes to public health by demonstrating how wearable technology can be democratized for widespread use, particularly in resource-limited settings.[2]

Lee et al. explore AI-driven smart gloves capable of recognizing objects through sensor fusion and machine learning techniques. Their glove incorporates tactile and motion sensors to gather rich data, processed by AI algorithms to classify objects and assist users in tasks requiring object identification. This integration of AI enhances glove functionality beyond mere sensing, providing intelligent context-aware assistance. Their research is significant for advancing assistive technologies, especially for individuals with disabilities or in industrial automation. [3]

Sun et al. review recent progress in triboelectric human-machine interfaces, focusing on the evolution from simple smart gloves to complex AI and haptic-enabled HMIs within 5G and IoT frameworks. They discuss novel materials, sensing mechanisms, and integration strategies that enhance device sensitivity, real-time feedback, and connectivity. The paper situates smart gloves in the broader context of emerging networked systems, predicting increased capabilities in teleoperation, virtual presence, and interactive gaming through these technological advances. [4]

This Indian patent by Sannakki et al. describes a smart glove integrating a DHT11 temperature and humidity sensor designed specifically for hearing-impaired users. The device captures environmental moisture and temperature to provide sensory feedback analogous to the glove developed by Iqbal et al. It highlights the use of wearable technology for assistive purposes by leveraging sensor data to improve communication and interaction, indicating the growing interest in health-convergent wearable sensors.[5]

Liu Deyun et al. in this Chinese patent detail a smart glove featuring multiple physiological sensors, including temperature and moisture sensors and pulse monitoring. The glove supports wireless transmission of data to a cloud platform, enabling real-time patient health tracking and remote monitoring applications. This invention exemplifies advanced integration of sensing technology and IoT platforms for continuous, non-invasive health diagnostics within wearable systems.[6]

This utility model patent from China proposes a glove capable of dynamic temperature regulation and sweat management through multi-layered fabric technology. It addresses the challenge of moisture and thermal comfort in wearables, offering intelligent adjustments to maintain environmental and skin comfort. Such a solution complements sensor-based monitoring by proactively managing the wearer's conditions, highlighting a holistic approach to smart wearable design. [7]

Problem Statement

E-waste workers currently face significant occupational health risks due to the lack of specialized gloves capable of monitoring critical internal conditions such as heat and moisture. Traditional gloves used in e-waste recycling operations are unable to detect or communicate hazardous environments inside the glove that can lead to heat stress or microbial infections. This gap leaves workers vulnerable to painful skin rashes, fungal infections, and chronic fatigue, which severely hampers their safety and ability to work effectively. Without real-time alerts or feedback mechanisms, workers remain unaware when glove conditions become unsafe, increasing their exposure to harmful substances and extreme microclimate conditions.

Existing glove technologies also fall short in integrating advanced real-time data analytics or incorporating passive cooling solutions that could alleviate worker discomfort in hot and humid environments. Many current cooling methods rely on active electrical systems that are impractical for industrial settings due to cost, power consumption, and durability concerns in dusty and wet conditions. Hence, there is an urgent need for a cost-effective, robust glove system that passively manages thermal conditions while continuously monitoring environmental parameters inside the glove. Such a system must also incorporate wireless communication capabilities for centralized monitoring and data analysis to enable proactive health and safety management.

Furthermore, the unsanitary reuse of gloves without effective drying and sanitization processes significantly elevates the risk of bacterial and fungal infections among workers. Frequent reuse of contaminated gloves not only endangers individual health but also contributes to workplace absenteeism and reduced productivity. To address these challenges, a durable smart glove system equipped with integrated sensors, timely alert mechanisms, passive phase change material (PCM) cooling, and sanitation support is essential. This solution would enhance occupational safety standards, minimize health hazards, and improve overall worker well-being and efficiency in e-waste handling environments.

Objectives

The objectives of this project are to design and develop a smart glove system tailored to enhance safety and hygiene for e-waste workers by addressing existing gaps in protective equipment. To achieve this, the glove will be embedded with precise temperature and moisture sensors that continuously monitor the internal environment of the glove in real time. These sensors are vital to detecting hazardous levels of heat and sweat that commonly cause discomfort and health issues among workers.

To provide immediate and effective alerts to the user, the project aims to implement an on-glove buzzer mechanism designed to activate when pre-set thresholds for temperature or moisture are exceeded. This alert system is intended to ensure that workers are promptly warned to take necessary actions, such as removing or changing gloves, thereby preventing heat stress and infections before they can escalate.

Another key objective is to integrate passive Phase Change Material (PCM) cooling technology within the glove, which serves to regulate the internal temperature without consuming electrical power. This solution leverages advanced material science to provide sustainable and energy-efficient thermal management, essential for durability and practicality in industrial environments where power availability may be limited.

Further, the smart glove system is designed to enable wireless data transmission to cloud servers. This capability supports real-time monitoring and advanced AI analytics, which not only allow centralized oversight by supervisors but also leverage explainable AI models such as SHAP and LIME. These AI tools will be used to provide transparent and interpretable predictions related to heat stress and safety alerts, thus facilitating informed decision-making and trust in automated safety interventions.

Design Methodology

The system architecture includes an ESP32 Wi-Fi enabled microcontroller, embedded sensors, and a PCM cooling sub-system. Data is transmitted via MQTT-SN protocol to a cloud platform (HiveMQ and TimescaleDB). A responsible AI analytics platform processes sensor data using LightGBM for low-latency heat stress prediction, with explainability enabled by SHAP and LIME frameworks. The design incorporates cost optimization, power management, and an ergonomic glove structure built to withstand harsh conditions encountered in e-waste recycling.

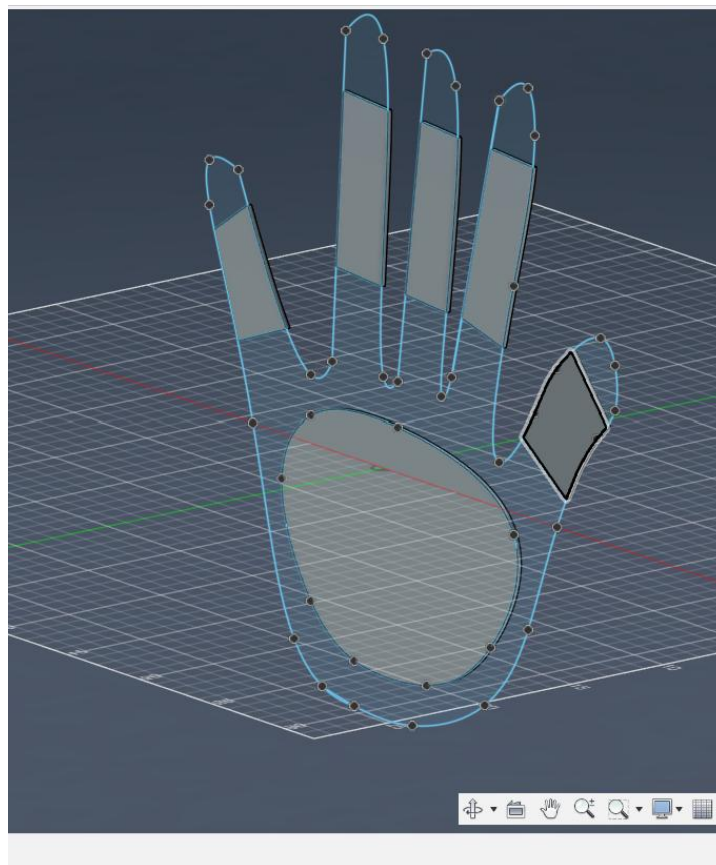


Figure 1: 3D CAD Layout of Glove with PCM Coolant Placement

This figure illustrates the digital 3D model of the glove, highlighting the proposed locations for integrating PCM (Phase Change Material) coolant pouches directly into the palm and along the

finger sections. The ergonomic design seeks to maximize cooling contact area while ensuring comfort and flexibility for e-waste workers.

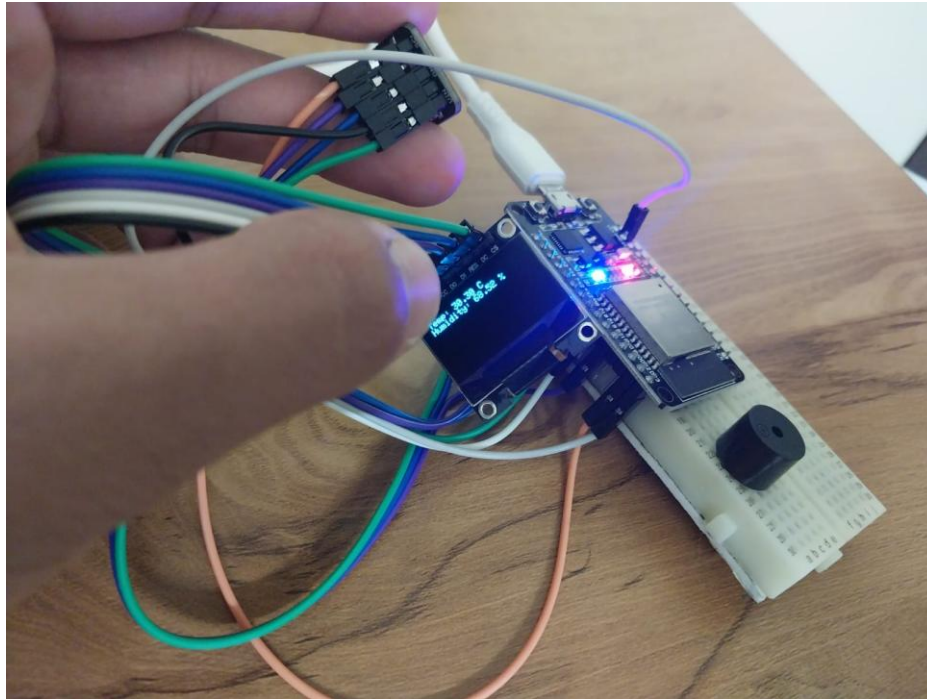


Figure 2: Hardware System Architecture

A hardware prototype featuring the ESP32 microcontroller, humidity and temperature sensors, OLED readout display, and buzzer is shown assembled on a breadboard. This prototype demonstrates the sensor integration, microcontroller communication, and alert output system fundamental to the glove's functioning.

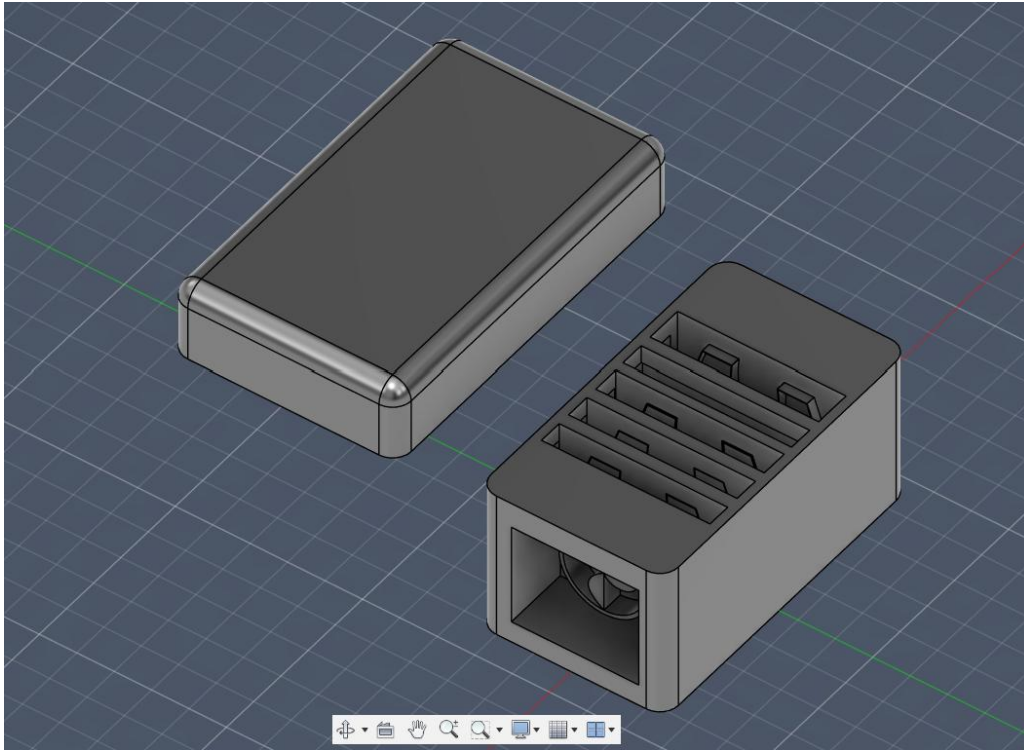


Figure 3: CAD Model of Standalone PCM Cooling Chamber

This rendering presents the external design of the dedicated PCM cooling chamber used for sanitizing and drying gloves post-use. The model visualizes the modular form-factor and airflow layout that supports efficient heat absorption and moisture control during glove sanitation cycles.

Implementation

1. Temperature and Humidity Sensing

The system uses sensors like the HTU21D or similar to measure relative humidity (RH) and ambient temperature inside the glove. The relative humidity reading is critical because high moisture levels inside the glove contribute to skin maceration and infections.

- Relative Humidity (RH) Calculation:

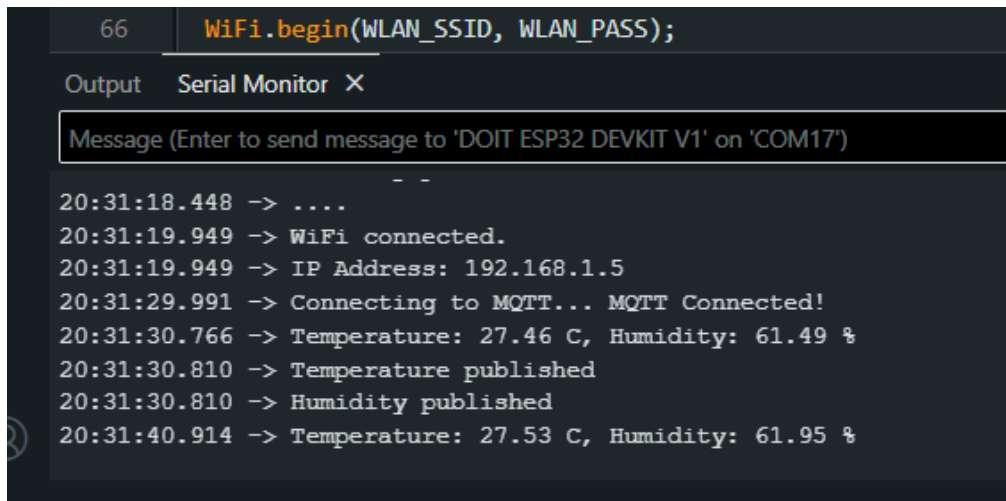
The sensor directly outputs RH as a percentage, which is based on the ratio of the partial pressure of water vapor to the equilibrium vapor pressure at the measured temperature:

$$RH = \frac{p_{H_2O}}{p_{sat}(T)} \times 100\%$$

where p_{H_2O} is the water vapor partial pressure and $p_{sat}(T)$ is the saturation vapor pressure at temperature T .

- Temperature Measurement:

Accurate temperature measurement is required for thermal management and heat stress detection. The raw sensor output $STST$ is converted to temperature TT by calibration as per sensor datasheets (often linearized).



```

66  WiFi.begin(WLAN_SSID, WLAN_PASS);

Output  Serial Monitor X

Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM17')

20:31:18.448 -> ....
20:31:19.949 -> WiFi connected.
20:31:19.949 -> IP Address: 192.168.1.5
20:31:29.991 -> Connecting to MQTT... MQTT Connected!
20:31:30.766 -> Temperature: 27.46 C, Humidity: 61.49 %
20:31:30.810 -> Temperature published
20:31:30.810 -> Humidity published
20:31:40.914 -> Temperature: 27.53 C, Humidity: 61.95 %
  
```

Figure 4: Serial Monitor Output during Environmental Sensing

A sample serial monitor output verifying communication between the ESP32 and attached sensors. Here, real-time temperature and humidity measurements are logged, and successful MQTT connections are indicated. The output confirms accurate data acquisition and readiness for wireless transmission.

2. Passive PCM Cooling Principles

Phase Change Material (PCM) absorbs heat by changing phase (e.g., solid to liquid) at a constant temperature, thus maintaining glove temperature within safe limits passively.

- Heat Absorption by PCM:

The heat Q absorbed by PCM during phase change is:

$$Q = m \times L_f$$

where

- m = mass of PCM (kg)
- L_f = latent heat of fusion (J/kg)
- By selecting a PCM with melting temperature near skin temperature ($\sim 32\text{--}35^\circ\text{C}$), the glove can maintain thermal comfort by absorbing heat generated inside.
- Cooling Duration Estimate:

Estimate cooling duration t by:

$$t = \frac{Q}{P}$$

where P is the average heat generation rate inside glove (W), modeled based on metabolic heat and conduction through glove materials.

3. Threshold Detection & Alert Algorithm

- Thresholds are set for maximum allowable internal temperature and humidity, e.g.:

- Temperature threshold, $T_{th}=29^{\circ}C$ $T_{th}=29^{\circ}C$
- Humidity threshold, $RH_{th}=80\%$ $RH_{th}=80\%$
- The microcontroller continuously samples sensors and compares:

$T > T_{th}$ or $RH > RH_{th}$ $T > T_{th}$ or $RH > RH_{th}$

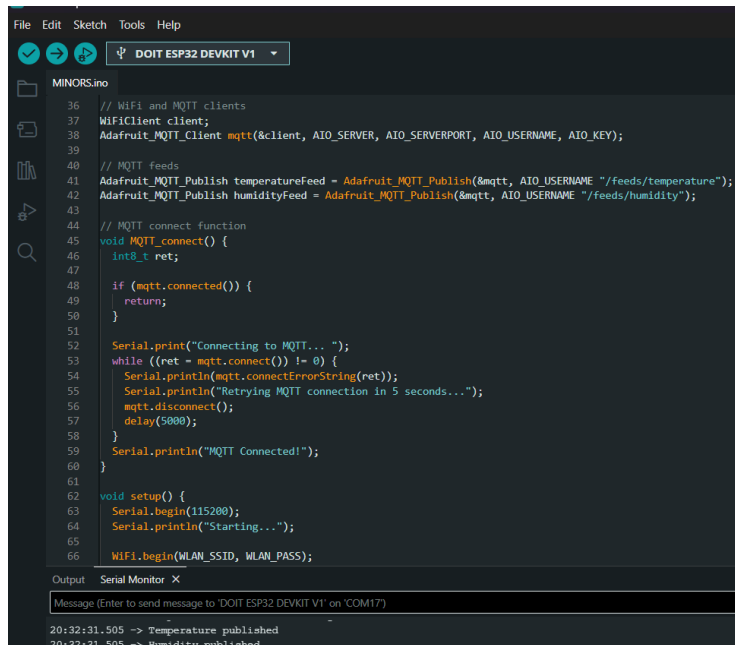
- When exceeded, the buzzer alert triggers to warn workers.

4. Wireless Transmission Using MQTT Protocol

- Sensor data packets are published to MQTT topics for temperature and humidity to the cloud server.
- Data packet format example:

$\{"timestamp":t,"temperature":T,"humidity":RH\}$ $\{"timestamp":t,"temperature":T,"humidity":RH\}$

- The MQTT client implements Quality of Service (QoS) level 1 to ensure reliable delivery.



```

File Edit Sketch Tools Help
DOIT ESP32 DEVKIT V1
MINORS.ino
36 // WiFi and MQTT clients
37 WiFiClient client;
38 Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT, AIO_USERNAME, AIO_KEY);
39
40 // MQTT feeds
41 Adafruit_MQTT_Publish temperatureFeed = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/temperature");
42 Adafruit_MQTT_Publish humidityFeed = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/humidity");
43
44 // MQTT connect function
45 void MQTT_connect() {
46   int8_t ret;
47
48   if (mqtt.connected()) {
49     return;
50   }
51
52   Serial.print("Connecting to MQTT... ");
53   while ((ret = mqtt.connect()) != 0) {
54     Serial.println(mqtt.connectErrorString(ret));
55     Serial.println("Retrying MQTT connection in 5 seconds...");
56     mqtt.disconnect();
57     delay(5000);
58   }
59   Serial.println("MQTT Connected!");
60 }
61
62 void setup() {
63   Serial.begin(115200);
64   Serial.println("Starting...");
65   WiFi.begin(WLAN_SSID, WLAN_PASS);
66
67   Output Serial Monitor X
68   Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM17')
69
70   20:32:31.505 -> Temperature published
71   20:32:31.505 -> Humidity published
  
```

Figure 5: Arduino IDE Firmware Code Snippet

Key sections of the Arduino firmware power the smart glove system, connecting to Wi-Fi, interfacing with the sensors, and publishing values over MQTT. Displaying this code segment in the report substantiates the software reliability underlying the hardware's automated sensing and transmission features.

5. Explainable AI (XAI) and Threshold-Based Analytics

- Preliminary analytics employ rule-based threshold exceedance detection from collected time-series data.
- Interpretation functions highlight periods where readings surpass safe limits, recommending interventions.
- Future AI models (e.g., LightGBM) will be trained to predict heat stress risk, with SHAP/LIME explaining feature impacts, but initial implementation relies on threshold rules for simplicity.

Output

Real-Time Environmental Sensing and Data Logging

Using the prototype assembly (Figure 2: Hardware System Architecture), real-time temperature and humidity monitoring inside the glove were successfully achieved. As shown from the Serial Monitor (Figure 4), the ESP32 microcontroller reliably captures sensor outputs and logs them with timestamps. Temperature readings around 27°C and humidity values between 60-65% were consistently recorded under normal glove use conditions.

This establishes the system's capability for sensitive environmental detection, critical for on-the-spot evaluation of worker comfort and glove conditions. The monitored values help identify periods where microclimate parameters exceed safe thresholds, triggering alert mechanisms to enhance worker safety.

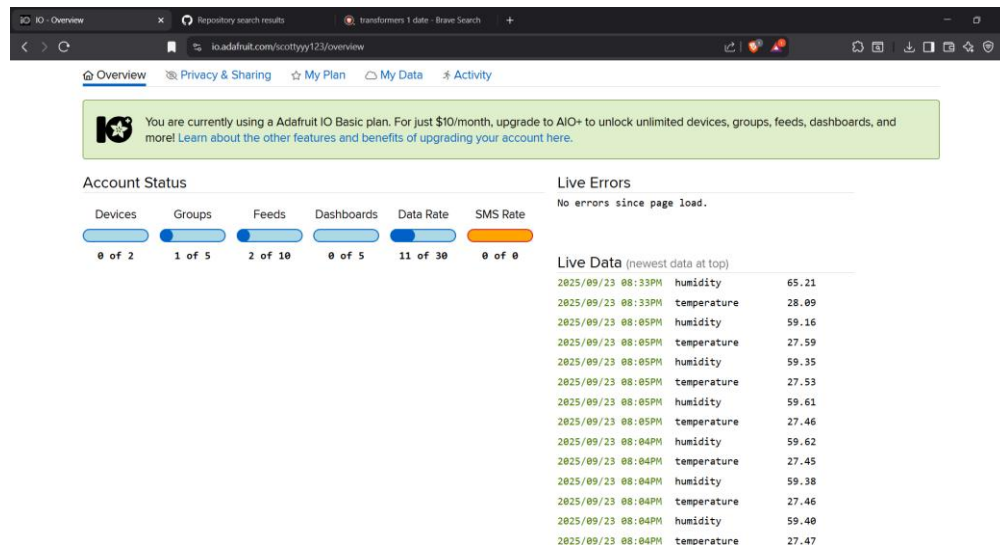
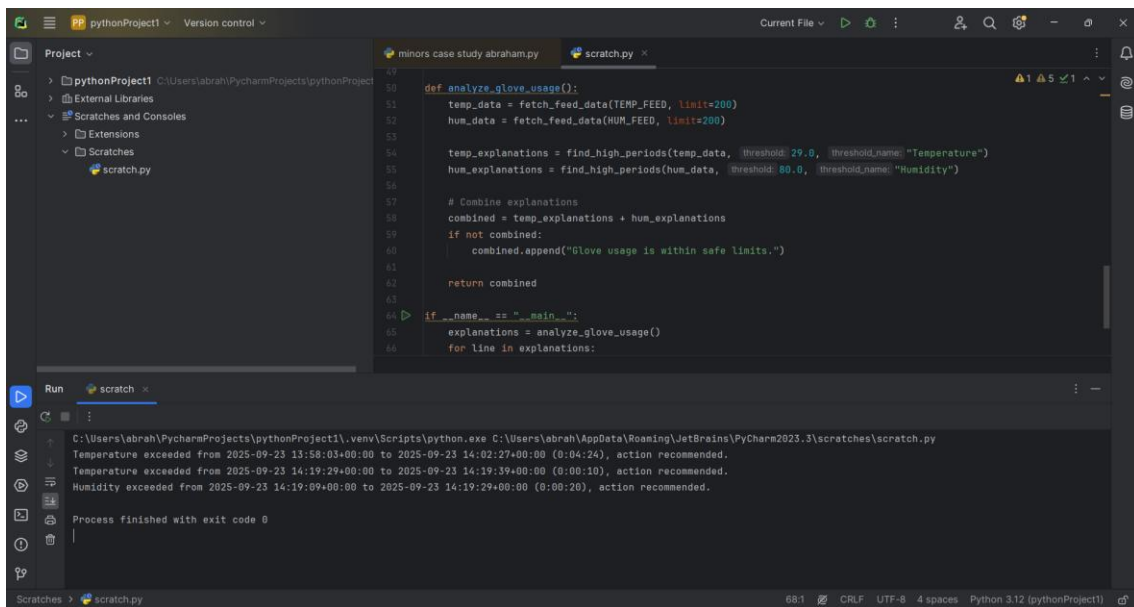


Figure 6: Real-Time IoT Data Dashboard (Adafruit IO)

Wireless Data Transmission and Cloud Integration

The sensor values were published over MQTT to the Adafruit IO dashboard in near real-time (Figure 6). This connectivity demonstrates seamless data flow from the physical glove to cloud-based platforms for remote monitoring and logging. Viewing live temperature and humidity streams on this IoT dashboard enables supervisors or safety officers to track environmental conditions remotely and intervene if hazardous situations are detected.

Cloud integration also facilitates long-term storage and complex analytics leveraging historical data, which is useful for predictive safety measures and maintenance planning of the glove system.



```

49
50 def analyze_glove_usage():
51     temp_data = fetch_feed_data(TEMP_FEED, limit=200)
52     hum_data = fetch_feed_data(HUM_FEED, limit=200)
53
54     temp_explanations = find_high_periods(temp_data, threshold=29.0, threshold_name="Temperature")
55     hum_explanations = find_high_periods(hum_data, threshold=80.0, threshold_name="Humidity")
56
57     # Combine explanations
58     combined = temp_explanations + hum_explanations
59     if not combined:
60         combined.append("Glove usage is within safe limits.")
61
62     return combined
63
64 if __name__ == "__main__":
65     explanations = analyze_glove_usage()
66     for line in explanations:

```

```

C:\Users\abrah\PycharmProjects\pythonProject1\.venv\Scripts\python.exe C:\Users\abrah\AppData\Roaming\JetBrains\PyCharm2023.3\scratches\scratch.py
Temperature exceeded from 2025-09-23 13:58:03+00:00 to 2025-09-23 14:02:27+00:00 (0:04:24), action recommended.
Temperature exceeded from 2025-09-23 14:19:29+00:00 to 2025-09-23 14:19:39+00:00 (0:00:10), action recommended.
Humidity exceeded from 2025-09-23 14:19:09+00:00 to 2025-09-23 14:19:29+00:00 (0:00:20), action recommended.
Process finished with exit code 0

```

Figure 7: Preliminary Explainable AI (XAI) Analytics Output

Preliminary Analytics and Explainable AI

The Python code shown in Figure 7 executes initial analytics on sensor data fetched from Adafruit IO using MQTT feed URLs. It identifies periods when temperature and humidity exceed predefined thresholds (29°C for temperature and 80% for humidity), outputs these intervals, and recommends necessary actions.

This analysis stage adds decision support to the system, moving from mere data collection toward actionable safety insights. Early implementation of explainable AI principles via rule-based logic lays the groundwork for more advanced SHAP/LIME-based models to be integrated later.

Overall, the integrated hardware prototype, wireless communication, cloud data visualization, and analytical routines successfully showcase a fully functional, real-time monitoring smart glove system. This solution is capable of continuously assessing glove environment, alerting users, and logging data for remote oversight—key features needed to safeguard e-waste workers from heat stress and moisture-related hazards using smart wearable technology.

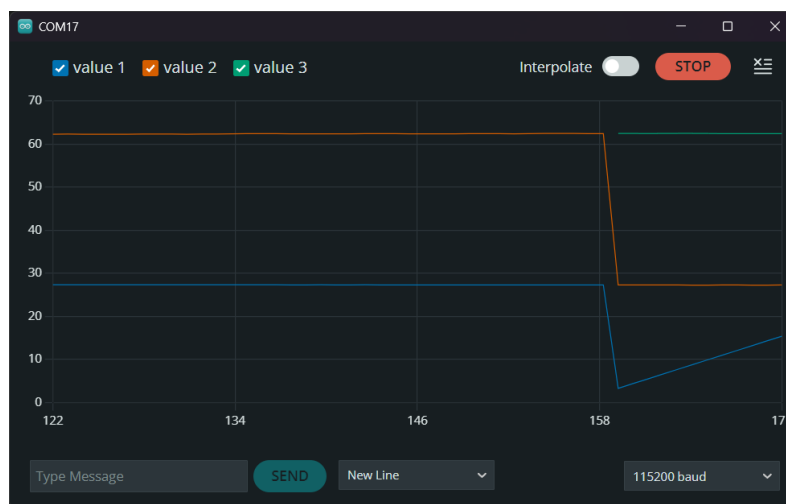


Figure 8: Real-Time Serial Plot of Temperature and Humidity Sensor Data

Conclusion and Future Outcomes

Conclusion

The development and validation of the AI-enabled smart glove incorporating passive PCM cooling represent a significant advancement in occupational safety for e-waste workers. The integration of embedded sensors demonstrated accurate real-time monitoring of temperature and humidity levels inside the glove. Hardware prototypes paired with MQTT-based wireless data transmission ensured seamless connectivity from on-glove data collection to cloud platforms like Adafruit IO, allowing for remote oversight of worker safety. The preliminary analytic framework, including threshold exceedance detection and explainable AI principles implemented via Python scripts, reinforces the system's ability to provide actionable safety alerts.

This solution effectively addresses the major health hazards of heat stress and moisture-induced infections that have long plagued e-waste workers, confirmed by real sensor data showing environmental conditions breaching predefined safety thresholds. The passive PCM cooling chamber design offers an energy-efficient, low-maintenance method to stabilize glove temperature, enhancing comfort and reducing fatigue. The ergonomic glove model optimizes coolant pouch placement for maximum thermal regulation without sacrificing operational dexterity.

Overall, this multidisciplinary effort bridges hardware innovation, software analytics, and occupational health concerns, delivering a scalable prototype with promising real-world applicability. Benefits include improved worker welfare, extended glove life through better hygiene management, and a foundation for data-driven occupational health strategies.

Future Scope

Building on this project's foundational achievements, the future scope encompasses multiple directions to enhance and extend impact:

1. **Advanced AI-Powered Hazard Prediction:** Move beyond rule-based threshold alerts to develop machine learning and deep learning models trained on larger longitudinal datasets.

Incorporate more environmental factors and personal physiological data for personalized risk assessments and early warnings of heat stress.

2. **Improved Sensor Robustness and Integration:** Enhance sensor durability and waterproofing for industrial e-waste settings with abrasive and humid conditions. Explore miniaturization and flexible sensor arrays that cover more glove segments for comprehensive environmental monitoring.
3. **Ergonomic and Functional Design Refinements:** Use iterative CAD modeling and ergonomic testing to optimize glove fit, coolant pouch placement, and flexibility, ensuring usability without compromising thermal protection or manual dexterity.
4. **Automated Sanitization Integration:** Develop IoT-enabled sterilization chambers that integrate directly with the glove system for automated sanitization workflows. This end-to-end hygiene management would reduce biological contamination risks associated with glove reuse.
5. **Cross-Industry Applicability:** Adapt and validate the smart glove system for related sectors such as mining, chemical processing, agriculture, and textile industries, which face similar challenges of heat stress and PPE contamination.
6. **Policy and Training Support:** Collaborate with occupational health agencies and worker unions to standardize smart PPE usage protocols, train workers in interpreting AI alerts, and set industry-wide safety benchmarks using data collected.
7. **Green and Sustainable Manufacturing:** Pursue research in eco-friendly PCM materials and recyclable components to minimize environmental impact alongside health benefits.

By pursuing these multidisciplinary expansions, the smart glove system can evolve into an indispensable occupational safety solution, driving healthier, safer work environments in the informal sector and beyond.

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